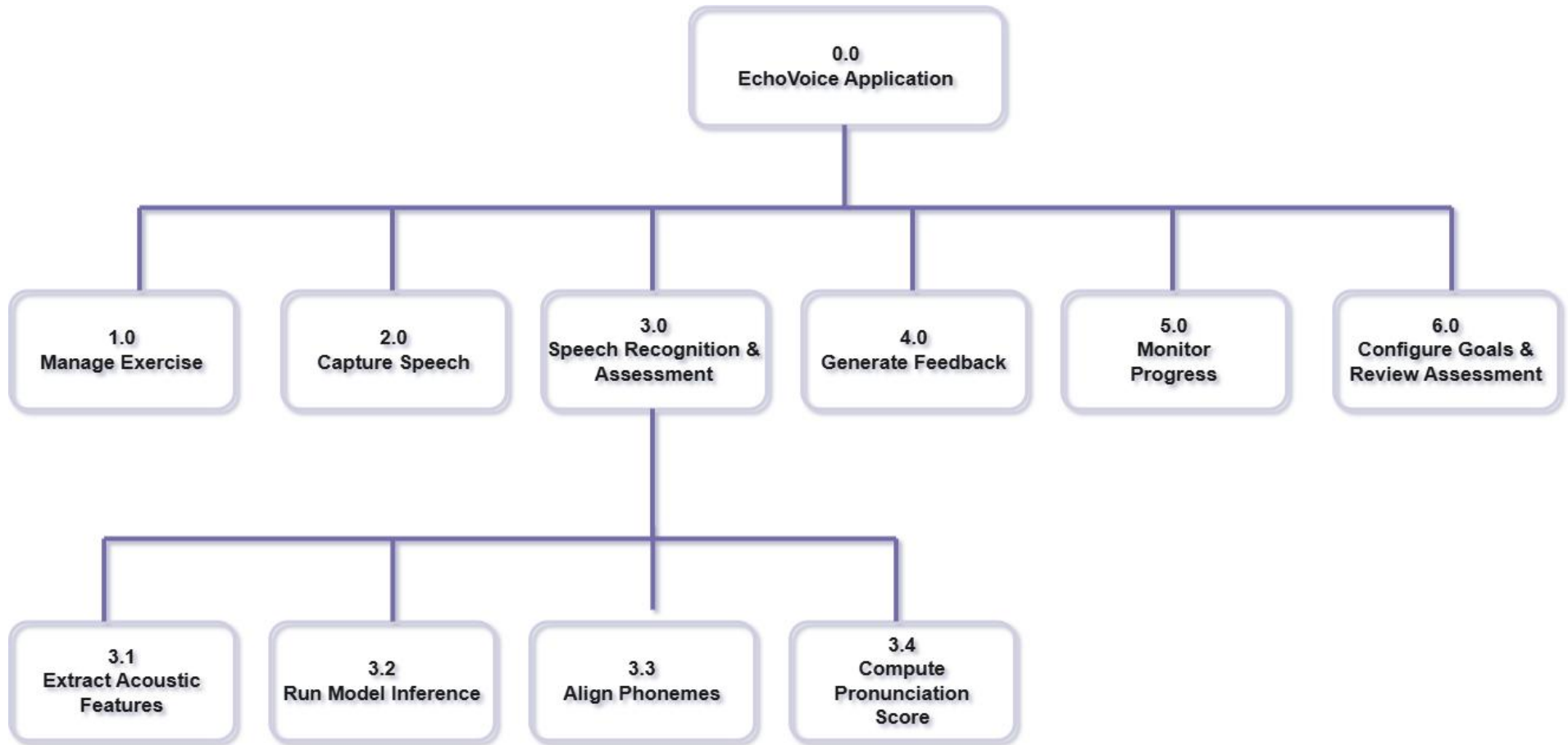


EchoVoice

HIPO DIAGRAM — HIERARCHY CHART (VTOC)



EchoVoice — HIPO: IPO Charts

On-Device Speech Recognition and Pronunciation Assessment App for Listen-and-Repeat Speech Therapy

0.0 EchoVoice Application (Overview)

Top-level module — corresponds to the Level 0 context DFD.

INPUT	PROCESS	OUTPUT
• Speech Input (Learner) • Exercise Selection (Caregiver) • Goal Configuration & Assessment Oversight (SLP)	• Coordinate exercise delivery, speech capture, on-device assessment, and reporting across all sub-modules	• Visual & Auditory Prompts → Learner • Session Progress Summary → Caregiver • Longitudinal Progress Reports → SLP

Level 1 Modules

1.0 Manage Exercise

Selects the next exercise for the learner based on caregiver input and the goals configured by the SLP.

INPUT	PROCESS	OUTPUT
• Exercise Selection (Caregiver) • Active Goals (D2 Goal Config Store)	• Match caregiver selection against active therapy goals • Retrieve corresponding target word	• Target Word → 2.0

2.0 Capture Speech

Records the learner's spoken attempt at the target word for downstream assessment.

INPUT	PROCESS	OUTPUT
• Speech Input (Learner) • Target Word (1.0)	• Capture microphone audio • Encode as raw audio file	• Recorded Speech → 3.0

3.0 Speech Recognition & Assessment

Runs the on-device model pipeline to score pronunciation accuracy. Decomposed into 3.1–3.4.

INPUT	PROCESS	OUTPUT
Recorded Speech (2.0)	Extract features → infer phonemes → align → score	- Assessment Result → 4.0 - Assessment Record → D1

4.0 Generate Feedback

Translates the assessment result into feedback the learner can act on in the moment.

INPUT	PROCESS	OUTPUT
Assessment Result (3.0)	Compose real-time visual/auditory feedback for the learner's attempt	Visual & Auditory Prompts → Learner

5.0 Monitor Progress

Aggregates stored assessment history into two distinct reporting outputs for two different audiences.

INPUT	PROCESS	OUTPUT
Stored Progress (D1 Progress Database)	- Compile trends across sessions - Build longitudinal report for the SLP - Build session-level recap for the Caregiver	- Longitudinal Progress Report → SLP - Session Progress Summary → Caregiver

6.0 Configure Goals & Review Assessment

Lets the SLP set therapy goals and review assessment oversight data.

INPUT	PROCESS	OUTPUT
Goal Configuration & Assessment Oversight (SLP)	Validate and record updated therapy goals	Store Goal Configuration → D2

Level 2 Modules — Decomposition of 3.0

Each stage feeds the next in sequence: 3.1 → 3.2 → 3.3 → 3.4.

3.1 Extract Acoustic Features

INPUT	PROCESS	OUTPUT
Raw Audio File (.wav) (2.0)	- Convert waveform to spectrogram - Encode as model-ready tensors	Spectrogram Tensors → 3.2

3.2 Run Model Inference

INPUT	PROCESS	OUTPUT
Spectrogram Tensors (3.1)	Run on-device speech recognition model	Predicted Phoneme Sequence → 3.3

3.3 Align Phonemes

INPUT	PROCESS	OUTPUT
- Predicted Phoneme Sequence (3.2) - Target Phoneme Template (D3)	Align predicted vs. target phoneme sequences	Phoneme Error Matrix → 3.4

3.4 Compute Pronunciation Score

INPUT	PROCESS	OUTPUT
Phoneme Error Matrix (3.3)	Score pronunciation accuracy from error matrix	- Assessment Result → 4.0 - Assessment Record → D1